Dr. Abderraouf Maoudj



Robot learning · Multi-robot coordination · Resilient robotic systems

Postdoctoral Research Fellow & Coordinator of the Robotics Laboratory

Interdisciplinary Research Center for Intelligent Manufacturing and Robotics
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1,837 citations · h-index 14 · i10-index 16 (Google Scholar, November 2026)

Research Profile

My research develops embodied intelligence for robot arms, legged robots and robot teams: systems that learn from limited experience, adapt when their hardware or tasks change, and coordinate safely at industrial scale. Over more than a decade, I have carried out academic and industrial research in Algeria, Denmark and Saudi Arabia, working with partners including Siemens, Mobile Industrial Robots and Enabled Robotics. My work connects robot learning, control and mechatronic design to flexible manufacturing, smart logistics and autonomous inspection.

- **Embodied AI and world models:** I develop NeuroAct, a modular robot intelligence architecture that combines embodiment representations, hierarchical policies and persistent memory to support adaptation across bodies and tasks. Its shared model has generalised to robot bodies not encountered during training in simulation. My current work develops multi-agent world models that learn how individual and collective actions shape the environment, alongside vision-language-action policies that anticipate physical interactions and reason about their timing.
- **Robot learning for intelligent manufacturing:** As Principal Investigator of KFUPM Grant INMR2403, I established a dual-arm Meca500 skill-learning testbed that is now shared laboratory infrastructure in active use. Decentralised policies with graph attention coordinate the physical arms and extend to larger teams in simulation. This work connects reusable manipulation skills, inter-arm coordination and motion planning to flexible industrial automation.
- **Multi-robot autonomy for smart logistics and environmental monitoring:** I develop decentralised path finding and scheduling for warehouse fleets operating with limited communication, in collaboration with Enabled Robotics. These methods coordinate large fleets and handle continuous task arrivals. Safety-filtered multi-drone policies improve gas-source localisation over infotaxis and model-predictive-control baselines, supporting research towards coordinated environmental monitoring.
- **Bio-inspired learning and fault-tolerant control for legged robots:** Reinforcement learning coupled with central pattern generators enables a physical 18-DoF hexapod to keep walking despite multiple disabled legs, without falls in repeated hardware trials. I am extending this direction through gait learning from a single insect walk using window-level adversarial inverse reinforcement learning, and online fuzzy-neural actor-critic control for hexapod trajectory tracking. The open-source SPINDER platform supports reproducible locomotion research; as Co-Principal Investigator of KFUPM Grant INMR2632, I also develop modular bipedal hardware and actuation.

Research Appointments

2023–Present **Postdoctoral Research Fellow & Robotics Laboratory Coordinator**
IRC-IMR, King Fahd University of Petroleum & Minerals, Saudi Arabia
Learning-based manipulation and multi-arm planning; decentralized multi-agent learning; resilient legged control and cross-embodiment policies. PI, INMR2403; Co-PI, INMR2632. Graduate teaching and student supervision.

2021–2023 **Postdoctoral Researcher**
SDU Biorobotics, University of Southern Denmark
Decentralized multi-robot coordination, communication-constrained path finding and capacitated fleet scheduling for industrial logistics. Worked with Enabled Robotics on MiR–UR5 mobile manipulators.

2018–2020 **Senior Researcher, Robotics Division**
Centre de Développement des Technologies Avancées (CDTA), Algeria
AGV fleet control, intelligent scheduling, collaborative robot integration and Siemens PLC automation for Industry 4.0; engineering teaching and project supervision.

2014–2018 **Robotics Engineer, Robotics Division**
CDTA, Algeria
Robot development, including an industrial AGV and a remotely operated bathymetric survey boat; doctoral research and laboratory teaching at USTHB.

Education

2013–2018 **Ph.D., Process Control and Interactive Rendering**
Université des Sciences et de la Technologie Houari Boumediene (USTHB), Algeria
Distinction: *Très Honorable*. Dissertation: *Teleoperation of Heterogeneous Robots: Application to RobuTER/ULM Robots and AGVM6 Robots.* Advisor: Prof. Brahim Bouzouia.

2010–2012 **M.Sc., Electronics — System Analysis**
Mohammed-Seddik Benyahia University of Jijel, Algeria
Thesis: *Design and Control of a Microcontroller-Based Mobile Robot Prototype.*

2007–2010 **B.Sc., Electronics**
Mohammed-Seddik Benyahia University of Jijel, Algeria

Teaching, Supervision and Mentorship

Current graduate teaching

ICS 520 — Artificial Intelligence and Machine Learning for Robotics, KFUPM, Fall 2026. Autonomous drone applications spanning perception, planning, decision-making and learning.

Courses taught

- **Automation Laboratory (AUTO503)**, undergraduate, USTHB, 2014/2015 and 2015/2016; **Control Laboratory**, Master's in Electronics of Embedded Systems, 2017/2018; **Industrial Programmable Logic Controllers (API)**, Master's in Instrumentation, 2018/2019.
- **Further teaching:** Linear Control Systems; Introduction to Robotic Systems and ROS; Applied Reinforcement Learning for Robotic Manipulation; Embedded Systems for Automotive Engineering; Algorithms and C Programming.
- **Curriculum development:** graduate courses in Reinforcement Learning for Robotics and Sim-to-Real Embodied Robotics, integrating classical control, robot learning and projects on open robotic platforms.

Graduate and project supervision

Co-supervised doctoral students at KFUPM and supervised or co-supervised Master's theses and final-year engineering projects at KFUPM and CDTA/USTHB. Topics include manipulation, industrial automation, navigation, multi-robot coordination and legged locomotion, with outcomes in industrial demonstrations, working prototypes and open research platforms.

Research Funding and Leadership

- **Principal Investigator — KFUPM Grant INMR2403 (2023–2025).** *Advancing Robotics for Industry 4.0: Integrating Collaborative Robot Systems to Enhance Industrial Tasks.* Led the project and established the shared dual-arm Meca500 skill-learning testbed. Project record.
- **Co-Principal Investigator — KFUPM Grant INMR2632 (ongoing).** *Intelligent Bipedal Robotic Platform.* Open, modular hardware and control infrastructure for locomotion research.
- **Postdoctoral Researcher — Swarm Robotics for Industry 4.0 (2021–2023).** Independent Research Fund Denmark project at SDU, with Enabled Robotics and IRIDIA, Université libre de Bruxelles. Project record.

Research and Development Projects

Next-Generation Robotic Manipulation and Learning Systems

KFUPM · Ongoing · Postdoctoral Research Fellow

Learning-based manipulation for flexible industrial automation combines reinforcement learning, imitation learning and inter-arm attention. A decentralised graph-attention policy runs on physical Meca500 arms and scales from up to four training arms to ten arms in simulation. Current work develops multi-agent world models for sparse-reward coordination and vision-language-action policies, with the goal of reusing manipulation skills as production tasks change.

Laboratory demonstrations · Multi-arm planning video

NeuroAct: Transferable Robot Intelligence Across Bodies and Tasks

KFUPM · Ongoing · Postdoctoral Research Fellow

A modular robot intelligence architecture combines embodiment representations, hierarchical policies and persistent memory to reduce retraining across hardware and tasks. Its shared model has generalised to unseen simulated robot morphologies, providing a foundation for investigating how learned skills transfer between different bodies.

Bio-Inspired Fault-Tolerant Control for Legged Robots

KFUPM · Ongoing · Postdoctoral Research Fellow

Reinforcement learning modulates central pattern generators to maintain coordinated locomotion as joints and legs fail. A single policy trained with randomised fault injection was deployed directly on a physical 18-DoF hexapod without additional hardware training, sustaining walking without falls across repeated trials with multiple disabled legs. This work advances adaptable locomotion for inspection and field robotics; complementary research integrates terrain perception and gait adaptation for quadrupeds.

SPINDER demonstration · Open platform and simulator

Learning Hexapod Gaits from a Single Insect Walk

KFUPM · Ongoing research

Developing window-level adversarial inverse reinforcement learning to capture gait structure from a single insect walking demonstration. The approach studies coordinated movement over time windows, with the aim of transferring biological locomotion patterns into adaptable hexapod behaviour while reducing the need for hand-designed gait rewards.

Online Fuzzy-Neural Actor-Critic Control for Hexapod Trajectory Tracking

KFUPM · Ongoing research

Developing an adaptive controller that combines fuzzy-neural approximation with online actor-critic learning for hexapod trajectory tracking. The research targets continuous adaptation to uncertain robot dynamics and changing operating conditions, connecting learning-based control to precise, reliable legged motion.

Intelligent Bipedal Robotic Platform

KFUPM · Grant INMR2632 · Ongoing · Co-Principal Investigator

Developing open, modular hardware and actuation for locomotion, balance recovery, jumping and obstacle-negotiation research. Replaceable mechanical, sensing and control modules, standardised interfaces and synchronised logging support reproducible experiments and comparison of classical and learning-based controllers.

Collaborative Robotics for Industry 4.0

KFUPM · Grant INMR2403 · 2023–2025 · Principal Investigator

Led the development of multi-cobot skill learning, coordination and motion planning for flexible production, alongside fleet coordination for storage and transport. Delivered a dual-arm Meca500 testbed with an integrated ROS stack that now supports shared research, teaching and experimental validation of manufacturing automation.

Project record

Swarm Robotics for Industry 4.0

SDU Biorobotics · 2021–2023 · Postdoctoral Researcher

Developed decentralised planning and capacitated scheduling for warehouse fleets operating with limited communication. Methods coordinate at least 1,000 robots in simulation and support lifelong operation with continuous task arrivals. Collaboration with Enabled Robotics, together with open planners and benchmark datasets, connects scalable coordination research to practical industrial logistics.

Project · ANTS planner · Benchmark dataset · Video

Industry 4.0 Platform Development

CDTA, with Siemens Algeria · 2016–2020 · Robotics Engineer / Senior Researcher

Integrated KUKA LBR iiwa collaborative assembly, RFID part tracking, Siemens PLCs and AGV navigation into a cyber-physical manufacturing platform with Siemens Algeria. The system connects robot autonomy, traceability and production control for coordinated industrial operations.

System demonstration

Aquatic Robot for Bathymetry

CDTA · 2016–2018 · Robotics Engineer

Designed the electronics and control of a remotely operated survey boat delivered to the Algerian National Hydrological Agency. The platform supports bathymetric measurement and environmental monitoring, translating robotic systems engineering into infrastructure for national water-resource assessment.

Survey boat technical record

Remote Control of Heterogeneous Robots

CDTA · 2014–2016 · Robotics Engineer / Doctoral Researcher

Developed distributed scheduling and control for heterogeneous industrial robot teams, combining reinforcement-learning path optimisation with fault-tolerant formation control. The work supports coordinated operation of mobile robots and manipulators across transport and production tasks.

Robots and Systems Designed and Built

- **SPINDER, open-source 18-DoF hexapod:** mechanical design, electronics, firmware, analytic inverse kinematics and hierarchical CPG control. Designs, models and code · Demonstration.
- **Dual-arm Meca500 testbed:** hardware integration and ROS stack for manipulation research and teaching.
- **Industrial AGV:** microcontroller-based motion control and lidar integration.
- **Industry 4.0 assembly cell:** KUKA LBR iiwa, Siemens UHF RFID and PLC integration. Video.
- **Bathymetric survey boat:** electronics and remote control for the Algerian National Hydrological Agency. Technical record.

Additional platform experience: UR5, UFactory xArm6 and Lite6, SO-ARM100, MiR mobile robots and Enabled Robotics mobile manipulators.

Technical Expertise

- **Robot learning:** PyTorch; reinforcement learning (PPO, SAC, TD3, DQN, offline RL); multi-agent learning; imitation learning; world models; transformers and vision–language–action models; sim-to-real transfer.
- **Planning and coordination:** multi-agent path finding; multi-arm motion planning; task allocation and scheduling; collision avoidance; mixed-integer optimization (CPLEX), genetic algorithms and vehicle routing.
- **Control and mechatronics:** kinematics and dynamics; nonlinear, fuzzy, sliding-mode and CPG control; control barrier functions; embedded motor control; STM32, Arduino and PIC; robot and sensor integration.
- **Software and automation:** Python, C/C++, MATLAB/Simulink, Java, C# and R; ROS 1/2, MoveIt; MuJoCo, Isaac Lab, PyBullet and CoppeliaSim; Siemens TIA Portal, Ladder and Grafcet; Git, Docker and Linux.

Invited Talks and Research Presentations

- **Reinforcement Learning in Robotics: From Mathematical Foundations to Real-World Applications.** Invited seminar, Electrical Engineering Department, KFUPM, November 2025.
- **Decentralized Multi-Robot Motion Planning and Coordination for Industrial Applications.** Invited seminar, University of Tartu, Estonia, September 2024.
- **Multi-Robot Motion Planning and Coordination in Industrial Environments.** Speaker and co-organizer, Fourth Annual Research Workshop, KFUPM, August 2024.
- Research demonstrations at the KFUPM Research Forum (October 2024); presented research at several international conferences in robotics, automation and industrial informatics.

Professional Service and Industry Engagement

- **Editorial and reviewing:** Topic Editor, *Frontiers in Robotics and AI*, "Human-Robot Interaction in Industrial Settings: New Challenges and Opportunities"; reviewer for robotics and AI journals and conferences.
- **Research infrastructure and open science:** Robotics Laboratory coordination; shared manipulation testbed; open SPINDER hardware and control code, ANTS planner and lifelong multi-agent benchmark datasets.
- **Industrial collaboration:** robotics and automation projects with Siemens Algeria, Enabled Robotics, Mobile Industrial Robots, Sonacome and the Algerian National Hydrological Agency.

Professional References

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Publications

Titles link to publisher records where available. Google Scholar.

Journal Articles — Published and Accepted

[J1] A. Aziz Khater, Y. Chen, A. Maoudj, A. Aouaichia, "Lyapunov-based adaptive path tracking control of wheeled mobile robots using a self-organizing recurrent Gaussian-gating functional-link fuzzy neural network," *Scientific Reports, published online 1 November 2026*.

[J2] B. Brahmi, A. Maoudj, J. Ghommam, "Nonlinear Terminal Model-Based Second-Order Sliding Mode Control for Robust Tracking in Robotic Exoskeletons," *International Journal of Robust and Nonlinear Control, Early View, 30 October 2026*.

[J3] A. Maoudj, B. Brahmi, A. A. Saputra, A. Aouaichia, M. Nouioua, N. Kubota, "Decentralized multi-drone gas source localization via belief-coupled MAPPO and barrier-filtered safety," *Scientific Reports*, accepted, 2026.

[J4] A. Maoudj, A. A. Saputra, B. Brahmi, M. Nouioua, "SPINDER: An open-source 18-DoF hexapod robot with hierarchical central pattern generator control and analytic inverse kinematics," *Scientific Reports*, vol. 16, art. 26147, 2026.

[J5] A. Maoudj, A. A. Saputra, B. Brahmi, N. Kubota, "A hierarchical perception-action framework for adaptive legged locomotion in unstructured environments," *IEEE Access*, vol. 14, pp. 74923–74932, 2026.

[J6] N. Djouambi, M. A. Yallese, M. Kaddeche, A. Maoudj, A. Khellaf, M. Nouioua, "Machine-Learning Surrogates for the Multi-Objective Optimisation of Cutting Parameters in Dry Turning of POM C GF25%," *The International Journal of Advanced Manufacturing Technology*, vol. 145, no. 5–6, pp. 3651–3670, 2026.

[J7] M. Nouioua, A. Laouissi, A. A. D. Sarhan, V. Swaminathan, A. Maoudj, H. M. Al-Qahtani, "AI-Driven Decision Support for Multi-Objective Optimization of Turning Parameters in Grey Cast Iron Machining," *Results in Engineering*, vol. 29, article 109805, 2026.

[J8] A. Mechta, A. Laouissi, E. Raouache, A. Maoudj, M. Nouioua, "Hybrid Numerical-Experimental Optimization of Resistance Spot Welding Parameters for Dissimilar S235/AISI 1075 Steel Sheets via Box–Behnken Response Surface Methodology," *JOM*, vol. 78, no. 11, pp. 10785–10805, 2026.

[J9] Z. Khan, A. Maoudj, S. Mekid, "From collaborative robots to Internet of Cobots: Innovations, applications, and emerging trends," *Journal of Ambient Intelligence and Smart Environments*, vol. 18, no. 1, pp. 41–67, 2026.

[J10] M. Boumediene, A. Maoudj, A. L. Christensen, "HM-DRL: Enhancing multi-agent pathfinding with a heatmap-based heuristic for distributed deep reinforcement learning," *Applied Intelligence*, vol. 55, no. 12, art. 873, 2025.

[J11] U. Farhadi, H. Hess, A. Maoudj, A. L. Christensen, "Evolution of path costs for efficient decentralized multi-agent pathfinding," *Swarm and Evolutionary Computation*, vol. 93, art. 101833, 2025.

[J12] A. Maoudj, A. L. Christensen, "Improved decentralized cooperative multi-agent path finding for robots with limited communication," *Swarm Intelligence*, vol. 18, no. 2–3, pp. 167–185, 2024.

[J13] A. Hentout, A. Maoudj, A. Kouider, "Shortest path planning and efficient fuzzy logic control of mobile robots in indoor static and dynamic environments," *Romanian Journal of Information Science and Technology*, vol. 27, no. 1, pp. 21–36, 2024.

[J14] A. Maoudj, A. Kouider, A. L. Christensen, "The capacitated multi-AGV scheduling problem with conflicting products: Model and a decentralized multi-agent approach," *Robotics and Computer-Integrated Manufacturing*, vol. 81, art. 102514, 2023.

[J15] A. Hentout, A. Maoudj, M. Aouache, "A review of the literature on fuzzy-logic approaches for collision-free path planning of manipulator robots," *Artificial Intelligence Review*, vol. 56, no. 4, pp. 3369–3444, 2023.

[J16] A. Hentout, A. Maoudj, A. Kouider, M. Aouache, "Effective GA approach for municipal solid waste collection: Dynamic capacitated vehicle routing in smart cities," *International Journal of Artificial Intelligence and Soft Computing*, vol. 7, no. 4, pp. 329–352, 2022.

[J17] A. Maoudj, A. Hentout, "Optimal path planning approach based on Q-learning algorithm for mobile robots," *Applied Soft Computing*, vol. 97, art. 106796, 2020.

[J18] A. Hentout, A. Maoudj, N. Kaid-Youcef, D. Hebib, B. Bouzouia, "Distributed multi-agent bidding-based approach for the collaborative mapping of unknown indoor environments by a homogeneous mobile robot team," *Journal of Intelligent Systems*, vol. 29, no. 1, pp. 84–99, 2020 (online 2017).

[J19] A. Hentout, M. Aouache, A. Maoudj, I. Akli, "Human-robot interaction in industrial collaborative robotics: A literature review of the decade 2008–2017," *Advanced Robotics*, vol. 33, no. 15–16, pp. 764–799, 2019.

[J20] A. Maoudj, B. Bouzouia, A. Hentout, A. Kouider, R. Toumi, "Distributed multi-agent scheduling and control system for robotic flexible assembly cells," *Journal of Intelligent Manufacturing*, vol. 30, no. 4, pp. 1629–1644, 2019.

[J21] A. Hentout, A. Maoudj, M. Ouahal, A. Gasmi, "Fault-tolerant multi-agent scheme for leader/follower formation control of homogeneous mobile robot team," *International Journal of Systems, Control and Communications*, vol. 10, no. 2, pp. 95–125, 2019.

[J22] A. Hentout, A. Maoudj, D. Guir, S. Saighi, M. A. Harkat, M. Z. Hammouche, A. Bakdi, "Collision-free path planning for indoor mobile robots based on rapidly-exploring random trees and piecewise cubic Hermite interpolating polynomial," *International Journal of Imaging and Robotics*, vol. 19, no. 3, 2019.

[J23] A. Hentout, A. Maoudj, D. Yahiaoui, M. Aouache, "RRT-A*-BT approach for optimal collision-free path planning for mobile robots," *Algerian Journal of Signals and Systems*, vol. 4, no. 2, pp. 39–50, 2019.

[J24] A. Maoudj, A. Hentout, B. Bouzouia, R. Toumi, "On-line fault-tolerant fuzzy-based path planning and obstacles avoidance approach for manipulator robots," *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, vol. 26, no. 5, pp. 809–838, 2018.

[J25] A. Bakdi, A. Hentout, H. Boutami, A. Maoudj, O. Hachour, B. Bouzouia, "Optimal path planning and execution for mobile robots using genetic algorithm and adaptive fuzzy-logic control," *Robotics and Autonomous Systems*, vol. 89, pp. 95–109, 2017.

Refereed Conference Papers

[C1] A. Maoudj, A. Kharidege, B. Brahmi, A. A. Saputra, M. Nouioua, "An Adaptive Vision-Based Robotic System for Flexible Surface Polishing," *23rd International Conference on Informatics in Control, Automation and Robotics (ICINCO)*, vol. 2, pp. 1091–1098, 2026.

[C2] B. Brahmi, A. Maoudj, J. Ghommam, M. H. Rahman, "Hamiltonian-Based Human Intention Estimation for Compliant Exoskeleton Robot," *23rd International Conference on Informatics in Control, Automation and Robotics (ICINCO)*, vol. 1, pp. 495–502, 2026.

[C3] K. Amara, W. Dib, A. O. Djekoune, A. Maoudj, N. Zenati, "AquaRob: An unmanned surface drone for the coastal surveillance and environmental monitoring," *Proc. 4th International Conference on Embedded & Distributed Systems (EDiS), IEEE*, pp. 193–198, 2024.

[C4] A. Maoudj, S. Mekid, A. L. Christensen, "Lifelong MAPF for shipment preparation in stochastic warehouses," *Proc. IEEE 20th International Conference on Automation Science and Engineering (CASE)*, pp. 1006–1011, 2024.

[C5] A. Maoudj, A. L. Christensen, “Decentralized multi-agent path finding in dynamic warehouse environments,” *Proc. 21st International Conference on Advanced Robotics (ICAR), IEEE*, pp. 28–34, 2023.

[C6] A. Maoudj, A. L. Christensen, “Decentralized multi-agent path finding in warehouse environments for fleets of mobile robots with limited communication range,” *Swarm Intelligence: 13th International Conference (ANTS 2022), Springer LNCS vol. 13491*, pp. 104–116, 2022.

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[C8] A. Maoudj, A. Hentout, A. L. Christensen, A. Kouider, “A fast constructive path planning algorithm for mobile robot navigation,” *Proc. 26th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)*, pp. 1–8, 2021.

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[C10] A. Maoudj, A. Hentout, M. Bouhebila, C. Boubakir, H. Maoudj, “Design and control of an educational and research-oriented mobile robot,” *International Conference on Networking, Telecommunications, Biomedical Engineering and Applications (ICNTBA)*, 2019.

[C11] H. Maoudj, A. Maoudj, A. Hentout, M. Bouhebila, C. Boubakir, “Autonomous fuzzy-based navigation of mobile robots: Design, realization and control of a prototype,” *International Conference on Networking, Telecommunications, Biomedical Engineering and Applications (ICNTBA)*, 2019.

[C12] A. Hentout, A. Tiberkak, A. Maoudj, L. Berkat, A. Abeb, “Virtual pheromone-based approach for objects searching in RFID-based cyber-physical robotic systems,” *Proc. International Conference on Applied Smart Systems (ICASS)*, pp. 1–7, 2018.

[C13] A. Hentout, A. Beghni, A. B. Nourine, A. Maoudj, B. Bouzouia, “Mobile robot RBPF-SLAM with LMS sensor in indoor environments: Application on RobuTER robot,” *Proc. 5th International Conference on Electrical Engineering, Boumerdes (ICEE-B), IEEE*, 2017.

[C14] A. Hentout, A. Maoudj, B. Bouzouia, “A survey of development frameworks for robotics,” *Proc. 8th International Conference on Modelling, Identification and Control (ICMIC)*, pp. 67–72, 2016.

[C15] A. Maoudj, B. Bouzouia, A. Hentout, A. Kouider, R. Toumi, “Distributed multi-agent approach based on priority rules and genetic algorithm for tasks scheduling in multi-robot cells,” *Proc. IECON 2016, 42nd Annual Conference of the IEEE Industrial Electronics Society*, pp. 692–697, 2016.

[C16] A. Maoudj, B. Bouzouia, A. Hentout, R. Toumi, “Multi-agent approach for task allocation and scheduling in cooperative heterogeneous multi-robot team: Simulation results,” *Proc. IEEE 13th International Conference on Industrial Informatics (INDIN)*, pp. 179–184, 2015.

[C17] A. Maoudj, B. Bouzouia, A. Hentout, R. Toumi, “Multi-agent architecture for telerobotics of heterogeneous multi-robot cooperation through negotiated task allocation with constraints: Initial implementation,” *1re Journée d’Étude en Automatique et ses Applications (1JEAA14)*, pp. 38–43, 2014.

Editorial, Workshop Paper, Preprints and Theses

[O1] S. Bouraine, M. Ammi, K. Geihs, A. Hentout, A. Maoudj, F. Yacef, “Editorial: Human-robot interaction in industrial settings: new challenges and opportunities,” *Frontiers in Robotics and AI*, vol. 12, art. 1652426, 2025.

[O2] A. Hentout, M. Aouache, A. Maoudj, I. Akli, “Key challenges and open issues of industrial collaborative robotics,” *Workshop on Human-Robot Interaction: From Service to Industry (HRI-SI), held at IEEE RO-MAN*, 2018.

[O3] A. Kharidege, A. Maoudj, “Integration of a vision system with a robotic polishing system for machining variety of workpieces,” *Research Square preprint*, 2024.

[O4] A. Hentout, A. Maoudj, A. Kouider, “Efficient fuzzy-logic control of indoor mobile robots evolving in static and dynamic workspaces,” *Research Square preprint*, 2023.

[O5] A. Hentout, A. Maoudj, A. Kouider, “A literature review of service robotics: Challenges and open research issues,” *Workshop on Human-Robot Interaction: From Service to Industry (HRI-SI)*, 2022.

[O6] A. Hentout, I. Akli, A. Maoudj, A. Tiberkak, “Developing a distributed architecture for robots control in industrial RFID-based cyber-physical systems,” *technical report/preprint*, 2018.

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Journal Manuscripts — Under Review and Submitted

[U1] A. A. Saputra, A. Maoudj, M. L. Afakh, A. R. Ahmad, S. Mekid, N. Kubota, “Decentralized RL-CPG control for hexapod locomotion under simultaneous multi-leg failures,” *IEEE Transactions on Robotics*, under review, 2026.

[U2] A. Maoudj, S. Mekid, M. Nouioua, “ML-BiRRT: Machine-learning-enhanced bi-directional RRT for multi-arm robot motion planning and coordination,” *Robotics and Autonomous Systems*, under review, 2026.

[U3] A. Maoudj, A. A. Khater, B. Brahmi, A. Aouaichia, A. M. Sallam, A. A. Saputra, “Online fuzzy-neural actor-critic control for hexapod trajectory tracking,” *Applied Soft Computing*, under review, 2026.

[U4] M. Boumediene, A. L. Christensen, A. Maoudj, “MAPF-Relocate: Mitigating congestion through strategic agent relocation in multi-agent pathfinding,” *Journal of Intelligent & Robotic Systems*, under review, 2026.

[U5] A. Maoudj, A. A. Khater, B. Brahmi, A. Aouaichia, A. M. Sallam, A. A. Saputra, P. K. M. Parameswarappa, “Lyapunov-stable path tracking control of wheeled mobile robots using self-organizing recurrent probabilistic functional-link fuzzy neural network,” *Biomimetic Intelligence and Robotics*, under review, 2026.

[U6] B. Brahmi, A. Maoudj, N. Iqbal, M. H. Rahman, “Model-based terminal second-order sliding mode control with exact finite-time convergence for fault-tolerant exoskeleton robots,” *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, under review, 2026.

[U7] A. A. Khater, A. Aouaichia, A. Maoudj, Y. Chen, “An advanced recurrent probabilistic TSK neuro-fuzzy identifier for mobile robot system,” *Engineering Applications of Artificial Intelligence*, under review, 2026.

[U8] S. Alqahtani, A. Maoudj, S. Bashmal, “Learning-based kinematic modeling and disturbance compensation for a 3UPU parallel kinematic manipulator under base vibrations,” *submitted*, 2026.